

Arduino Nano Final Pin Plan (with W5500 + TM1637)

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■ Pin Assignments

Function	Pin(s)	Notes
Seed analog (random)	A0	Read once before reuse
Ethernet reset (ETH_RST)	A0	Set OUTPUT after seeding
Feedback inputs (7)	A1–A5, D0, D1	Only when <code>DEBUG_SERIAL == false</code>
Extra input (1)	A6	Analog-only, treat like A7
Pushbutton (Cycle Station)	A7	Analog-only
Relay outputs (6)	D2, D3, D4, D5, D6, D7	—
TM1637 Display	D8 = CLK, D9 = DIO	—
W5500 Ethernet	D10 (CS), D11 (MOSI), D12 (MISO), D13 (SCK)	Built-in LED also on D13 (don't drive)

■■ Important Wiring & Firmware Notes

A6 & A7 (Analog-only)

- No `pinMode()` or internal pullups.
- Use **external 10 kΩ resistor** (pull-up or pull-down).
- Example (pull-up): 10 kΩ from A7 → +5 V, button from A7 → GND.
- Threshold: pressed `<200`, released `>300`.

D0 & D1 (Digital inputs when not debugging)

- Configure as `INPUT_PULLUP`.
- Add **1–4.7 kΩ series resistor** between pin and GND via switch.
- Do **not** press switches on D0/D1 during uploads or Serial debugging.

A0 dual use (seed → ETH_RST)

1. `randomSeed(analogRead(A0) ^ micros());`
2. Then `pinMode(A0, OUTPUT)`.
3. Pulse LOW→HIGH (10–100 ms) to reset W5500.

Debounce

- Keep your existing **25 ms debounce** for all inputs.
- For A6/A7, debounce after analog thresholding.

Summary

This configuration ensures:

- Reliable Ethernet reset sequence (A0 shared safely)
- Clean digital feedback inputs (A1–A5, D0, D1)
- Analog-based pushbutton (A7) and extra input (A6)
- No SPI or TM1637 interference
- Safe upload/debugging behavior when Serial active